while SEM allows for the estimation of the combined effects of independent variable into concepts/ construct.

3) SEM includes measurement error into the model. A problem for path analyses using regression as its underlying analysis not adequately control for measurement error. SEM analyses do account for measurement error, therefore providing a better understanding of how good the theoretical model predicts actual behaviour.

17.13.2 Disadvantages of SEM

1) Most of the procedures that have been suggested involve nonstandard and complex model specifications that are challenging for the average user and thus susceptible to error. Indeed, errors have even been noted in the specifications developed by SEM specialists.

2) Because products of normally distributed observed and latent variables are themselves not normally distributed, standard errors and estimates of fit might not be accurate. This problem will be more severe to the extent that the latent exogenous variables used to form the product term are highly correlated.

3) If the latent variables that denote main effects are not normally distributed, the parameter estimates yielded by several procedures are not consistent.

4) Most of the methods proposed in the literature are applicable to a restricted class of measurement models.

5) Although a number of alternative procedures and options have been proposed, the selection of an optimal approach is made difficult by the absence of any one study or set of studies that compares all the viable alternatives that have been proposed to date under a variety of conditions.

6) Some of the more promising approaches are not easily available in conventional SEM software.

17.14 LET US SUM UP

SEM techniques are considered today to be a major component of applied multivariate statistical analysis and are used by education researchers, economists, marketing researchers, medical researchers, and variety of other social and behavioural scientists. SEM is a collection of statistical techniques that allow a set of relations between one or more independent variables(IVs), and one or more dependent variables (DVs). SEM is an attempt to model causal relations between variables by including all variables that are known to have some involvement in the process of interest. It can be viewed as a combination of factor analysis and regression or path analysis. Structural equation modeling (SEM) uses various types of models to depict relationships among observed variables, with the same basic goal of providing a quantitative test of a theoretical model hypothesized by the researcher. Two goals in SEM are (1) to understand the patterns of correlation/covariance among a set of variables and (2) to explain as much of their variance as possible with the model specified.

To discuss the history of structural equation modeling, it is better to explain the chronological order of following four models : regression, path, confirmatory factor, and structural equation models.

One of the easiest ways to communicate a structural equation model is to draw a diagram of it, referred to as *path diagram*, using special graphical notation. A path diagram is a form of graphical representation of a model under consideration. Such a diagram is equivalent to a set of equations defining a model (in addition to distributional and related assumptions), and is typically used as an alternative way of presenting a model pictorially. Path diagrams not only enhance the understanding of structural equation models and their communication among researchers with various backgrounds, but also substantially contribute to the creation of correct command files to fit and test models with specialized programs. Five steps are involed in SEM: Initial model conceptualization, Model estimation, Data-model fit assessment, Potential model modification and Reporting the Results. Most SEM analyses are conducted using one of the specialized SEM software programs. These include LISREL, EQS, Mplus, Amos and Mx. The SEM has certain advantages and disadvantages also.

17.15 KEY WORDS

SEM	:	Collection of statistical techniques that allow a set of relations between one or more independent variables (IVs), and one or more dependent variables (DVs).
Path Diagrams	:	Pictorial representation of a model.
Observed variables	:	Variables that are directly measured.
Unobserved variable	:	Latent variable in terms of multiple indicators that are in common with each other.
Residual or error terms	:	Can be associated with either observed variables or factors specified as outcome (dependent) variables.
Measurement model	:	The part of the model that relates the measured variables to the factors.
ξ (ksi)	:	Latent construct associated with observed x_i indicators.
η (eta)	:	Latent construct associated with observed y_i indicators.
δ (delta)	:	Error term associated with observed x_i indicators.
ε (epsilon)	:	Error term associated with observed x_i, y_i indicators.
ζ (zeta)	:	Error term associated with the formative construct.
Λij	:	Factor loading in the i-th observed indicator that is explained by the j-th latent construct.
Γij	:	Weight in the i-th observed indicator that is explained by the j-th latent construct.

Reflective indicator	:	The construct is the cause of the observed measures, so a variation in the construct leads to a variation in all its measures.
Formative or causal indicator	:	The direction of causality goes from the indicators (measures) to the construct and the error term is estimated at the construct level.
Model specification	:	Involves using all available relevant theory, research, and information to construct the theoretical model.
Model identification	:	A model is said to be identified if there is a unique numerical solution for each of the parameters in the model.
Underidentified	:	If one or more parameters cannot solved uniquely on the basis of the covariance matrix **S** (more unknown parameters than equations).
Just-identified	:	There is just enough information in the covariance matrix **S** to solve the parameters of the model (equally number of equations as unknown parameters).
Over-identified	:	More than one way to estimate the unknown parameters (more equations than unknowns).
Model estimation	:	Minimize the difference between the structured and unstructured estimated population covariance matrices.
Model evaluation	:	Assess the fit between observed data and the hypothesized model ideally operationalized as an evaluation of the degree of discrepancy between the true population covariance matrix and that implied by the model's structural and non structural parameters.
Model modification	:	Respecifying the model based on more relevant theory.
Nested models	:	Models that are subsets of one another.

17.16 REFERENCES

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